Apply the Sum and Difference Formulas for Sine and Cosine
Apply the Sum and Difference Formulas for Tangent

Sum and Difference Formulas

$$\sin(u+v)=\sin u\cos v+\cos u\sin v \qquad \cos(u+v)=\cos u\cos v-\sin u\sin v$$
$$\sin(u-v)=\sin u\cos v-\cos u\sin v \qquad \cos(u-v)=\cos u\cos v+\sin u\sin v$$

$$\tan(u+v)=\frac{\tan u+\tan v}{1-\tan u\tan v} \qquad \tan(u-v)=\frac{\tan u-\tan v}{1+\tan u\tan v}$$

Use an addition or subtraction formula to find the exact value. $45^\circ=\frac{\pi}{4}=\frac{3\pi}{12}$, $30^\circ=\frac{\pi}{6}=\frac{2\pi}{12}$, $60^\circ=\frac{\pi}{3}=\frac{4\pi}{12}$

1. $\sin 15^\circ \quad 15^\circ=45^\circ-30^\circ$

$$=\sin(45^\circ-30^\circ)=\sin 45^\circ\cos 30^\circ-\cos 45^\circ\sin 30^\circ$$
$$=\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2}-\frac{\sqrt{2}}{2}\cdot\frac{1}{2}=\frac{\sqrt{6}-\sqrt{2}}{4}$$

2. $\cos\frac{7\pi}{12} \quad \frac{7\pi}{12}=\frac{3\pi}{12}+\frac{4\pi}{12}=\frac{\pi}{4}+\frac{\pi}{3}$

$$=\cos\left(\frac{\pi}{4}+\frac{\pi}{3}\right)=\cos\frac{\pi}{4}\cos\frac{\pi}{3}-\sin\frac{\pi}{4}\sin\frac{\pi}{3}$$
$$=\frac{\sqrt{2}}{2}\cdot\frac{1}{2}-\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2}=\frac{\sqrt{2}-\sqrt{6}}{4}$$

3. $\tan 75^\circ \quad 75^\circ=45^\circ+30^\circ$

$$=\tan(45^\circ+30^\circ)=\frac{\tan 45^\circ+\tan 30^\circ}{1-\tan 45^\circ\tan 30^\circ}=\frac{1+\frac{\sqrt{3}}{3}}{1-\frac{\sqrt{3}}{3}}\cdot\frac{3}{3}$$
$$=\frac{(3+\sqrt{3})}{(3-\sqrt{3})}\frac{(3+\sqrt{3})}{(3+\sqrt{3})}=\frac{9+6\sqrt{3}+3}{9-3}=\frac{12+6\sqrt{3}}{6}=\frac{\cancel{6}(2+\sqrt{3})}{\cancel{6}}=2+\sqrt{3}$$

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4. $\sin\left(-\frac{\pi}{12}\right)$ $-\frac{\pi}{12}=\frac{3\pi}{12}-\frac{4\pi}{12}=\frac{\pi}{4}-\frac{\pi}{3}$ also note $\sin\left(-\frac{\pi}{12}\right)=-\sin\frac{\pi}{12}$

$$=\sin\left(\frac{\pi}{4}-\frac{\pi}{3}\right)=\sin\frac{\pi}{4}\cos\frac{\pi}{3}-\cos\frac{\pi}{4}\sin\frac{\pi}{3}$$

$$=\frac{\sqrt{2}}{2}\cdot\frac{1}{2}-\frac{\sqrt{2}}{2}\cdot\frac{\sqrt{3}}{2}=\frac{\sqrt{2}-\sqrt{6}}{4}$$

5. $\sin 65^\circ\cos 20^\circ-\cos 65^\circ\sin 20^\circ$

$$=\sin\left(65^\circ-20^\circ\right)=\sin 45^\circ=\frac{\sqrt{2}}{2}$$

$\sin u\cos v-\cos u\sin v=\sin(u-v)$

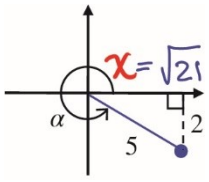
6. $\dfrac{\tan\frac{7\pi}{9}-\tan\frac{\pi}{9}}{1+\tan\frac{7\pi}{9}\tan\frac{\pi}{9}}=\tan\left(\frac{7\pi}{9}-\frac{\pi}{9}\right)=\tan\frac{6\pi}{9}=\tan\frac{2\pi}{3}=-\sqrt{3}$

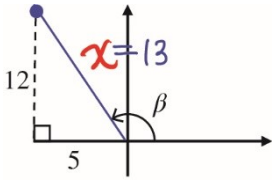
$\dfrac{\tan u-\tan v}{1+\tan u\tan v}=\tan(u-v)$

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7. Find the exact value of $\cos(\alpha-\beta)$.

$\sin\alpha=-\frac{2}{5}$ for α in Quadrant IV and $\tan\beta=-\frac{12}{5}$ for β in Quadrant II.





$$\left.\begin{aligned}x^2+2^2&=5^2\\x^2&=25-4\\x^2&=21\\x&=\sqrt{21}\end{aligned}\right\}\quad\begin{aligned}\sin\alpha&=-\frac{2}{5}\\\cos\alpha&=\frac{\sqrt{21}}{5}\end{aligned}$$

$$\left.\begin{aligned}12^2+5^2&=x^2\\144+25&=x^2\\169&=x^2\\x&=13\end{aligned}\right\}\quad\begin{aligned}\sin\beta&=\frac{12}{13}\\\cos\beta&=-\frac{5}{13}\end{aligned}$$

$$\cos(\alpha-\beta)=\cos\alpha\cos\beta+\sin\alpha\sin\beta$$

$$=\frac{\sqrt{21}}{5}\left(-\frac{5}{13}\right)+\left(-\frac{2}{5}\right)\frac{12}{13}$$

$$=\frac{-5\sqrt{21}-24}{65}$$

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Use the Sum and Difference Formulas to Verify Identities

Verify the identity.

8. $\sin\left(x-\frac{3\pi}{2}\right)=\cos x$

$$\text{LHS: } \sin\left(x-\frac{3\pi}{2}\right)=\sin x\cos\frac{3\pi}{2}-\cos x\sin\frac{3\pi}{2}$$
$$=\sin x\cdot 0-(-1)\cos x$$
$$=\cos x \quad \text{(RHS)}$$

9. $\cos(x+y)-\cos(x-y)=-2\sin x\sin y$

$$\text{LHS: } \cos x\cos y-\sin x\sin y-(\cos x\cos y+\sin x\sin y)$$
$$=\cos x\cos y-\sin x\sin y-\cos x\cos y-\sin x\sin y$$
$$=-2\sin x\sin y \quad \text{(RHS)}$$

10. $\tan\left(x+\frac{\pi}{4}\right)\tan\left(x-\frac{\pi}{4}\right)=-1$

$$\text{LHS: } \left(\frac{\tan x+\tan\frac{\pi}{4}}{1-\tan x\tan\frac{\pi}{4}}\right)\left(\frac{\tan x-\tan\frac{\pi}{4}}{1+\tan x\tan\frac{\pi}{4}}\right)$$
$$=\frac{\tan x+1}{1-\tan x}\cdot\frac{\tan x-1}{1+\tan x}=-1 \quad \text{(RHS)}$$

11. $\frac{\sin(x+y)}{\cos x\cos y}=\tan x+\tan y$

$$\text{LHS: } \frac{\sin x\cos y+\cos x\sin y}{\cos x\cos y}=\frac{\sin x\cos y}{\cos x\cos y}+\frac{\cos x\sin y}{\cos x\cos y}$$
$$=\tan x+\tan y \quad \text{(RHS)}$$

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Write a Sum of $A\sin x$ and $B\cos x$ as a Single Term

This formula is called a reduction formula because it "reduces" the sum of two trigonometric functions into one term.

For real numbers A, B, and x,

$$A\sin x + B\cos x = k\sin(x+\alpha)$$

where $k = \sqrt{A^2 + B^2}$, and α satisfies $\cos\alpha = \frac{A}{k}$ and $\sin\alpha = \frac{B}{k}$.

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12. Write $11\sin x + 60\cos x$ in the form $k\sin(x+\alpha)$ for $0 \le \alpha < 2\pi$. Round α to 4 decimal places.

$A = 11, B = 60$

$k = \sqrt{A^2 + B^2} = \sqrt{11^2 + 60^2} = \sqrt{3721} = 61$

$\cos\alpha = \frac{A}{k} = \frac{11}{61}, \sin\alpha = \frac{B}{k} = \frac{60}{61}$

$\alpha = \cos^{-1}\frac{11}{61} \approx 1.3895 \quad \cos\alpha > 0, \sin\alpha > 0 \rightarrow \text{QI}$

$11\sin x + 60\cos x \approx 61\sin(x + 1.3895)$

13. Check your answer to Exercise 12 by using the sum formula for sine.

$$\begin{aligned} 61\sin(x+1.3895) &\approx 61(\sin x\cos 1.3895 + \cos x\sin 1.3895) \\ &= 61\sin x\cos 1.3895 + 61\cos x\sin 1.3895 \\ &\approx 10.9986\sin x + 60.0003\cos x \\ &\approx 11\sin x + 60\cos x \end{aligned}$$

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